Formula 1

A Mathematical Analysis to Identify the Top Formula 1 Driver in History

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Contents

1. Introduction **3**

2. Dataset **4**

2.1 Source 4

2.2 Files 4

2.3 Tables 4

2.3.1 `circuits.csv` 4

2.3.2 `constructor_results.csv` 5

2.3.3 `constructor_standings.csv` 5

2.3.4 `constructors.csv` 5

2.3.5 `driver_standings.csv` 5

2.3.6 `drivers.csv` 6

2.3.7 `races.csv` 6

2.3.8 `results.csv` 6

2.3.9 `status.csv` 7

2.4 Entity Relationship Diagram Model 7

3. Data Transformation **8**

9. References **9**

DRAFT

1. Introduction

When posed the question, "Who is or was the best Formula 1 driver of all time?" an AI agent will likely first mention **Lewis Hamilton**, followed by **Michael Schumacher**, based primarily on their respective world championship titles. If asked for a list of top drivers, it might also include **Juan Manuel Fangio**, **Alain Prost**, **Max Verstappen**, and **Sebastian Vettel**, arranged alphabetically. **Fangio** stands out with five championships, while the others each have four.

Interestingly, many fans regard **Ayrton Senna**, with his three titles, as a superior driver compared to **Alain Prost**, despite the latter's four championships. This raises the question: how significant are titles in evaluating a driver's greatness? Should we consider alternative criteria for determining who truly excels?

If we delve deeper, **Juan Manuel Fangio** won five titles in the 1950s, during the formative years of the Formula 1 World Championship. His legacy is so profound that his name has become synonymous with exceptional driving. In French-speaking regions, the phrase "driving like Fangio" denotes flawless driving. A notable reference appears in *The Adventures of Tintin: The Calculus Affair* [1], where a character remarks, "This must be Fangio at the wheel" as a car skillfully avoids a net, reflecting Fangio's prowess during the height of his career. (In the English edition, it is translated as *"They must have a Jack Brabham at the wheel!"*)

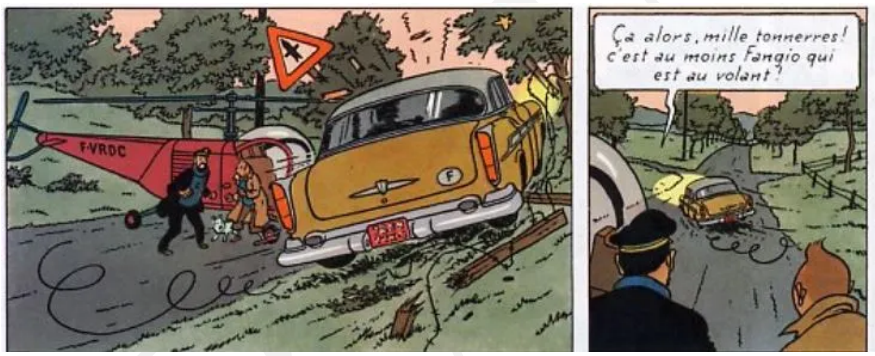

Figure 1: *"This must be Fangio at the wheel!"*

Complicating matters further, the rules of Formula 1 have evolved significantly over the years. The number of races has increased (from seven during Fangio's era to 22 today), and the points system has undergone various changes. For instance, the winner received 8 points until **1960**, then 9 points until **1990**, followed by 10 points until **2009**, and has been set at 25 points since **2010**. Additionally, until **1990**, only the best results counted towards the championship, which allowed Senna to secure the title in **1990** against Prost, despite the latter having more points overall; under modern rules, Prost would have been awarded the title instead.

Ultimately, it is evident that even the most skilled drivers depend significantly on their cars' performance. Both Alain Prost in **1991** and Lewis Hamilton in **2025** encountered substantial difficulties while driving for Ferrari, so much so that Prost was dismissed from his team after comparing his car to a "truck." This raises an important question: to what extent is a world title attributable to the car, and how much is a reflection of the driver's ability?

2. Dataset

2.1 Source

The Formula 1 data, including information on drivers, teams, qualifications, and results, is readily available online, such as on Wikipedia [2]. However, for research purposes, a well-maintained dataset may be more valuable. A suitable dataset can be found on Kaggle [3], although accessing it requires an account that some users may not possess. A publicly accessible version of this dataset is also available on GitHub Training Insights, but to guarantee its ongoing accessibility, I have uploaded a copy to my GitHub repository [4]. This redundancy will ensure that the dataset remains available even if the original site removes it.

2.2 Files

The dataset is organized as nineteen different files : 18 comma-separated value files and a json file. We will only use the 9 files below :

Dataset Files
circuits.csv
constructor_results.csv
constructor_standings.csv
constructors.csv
driver_standings.csv
drivers.csv
races.csv
results.csv
status.csv

2.3 Tables

2.3.1 `circuits.csv`

This file contains information about each circuit where a Formula 1 race has been held.

Column	Description
`circuitId`	Unique identifier for each circuit
`circuitRef`	A short reference name for the circuit
`name`	The official name of the circuit
`location`	The city where the circuit is located
`country`	The country where the circuit is located
`lat`	Latitude of the circuit
`lng`	Longitude of the circuit
`alt`	Altitude of the circuit in meters
`url`	URL for the circuit's Wikipedia page

DRAFT

2.3.2 `constructor_results.csv`

This file contains the results for each constructor in each race.

Column	Description
`constructorResultsId`	Unique identifier for each constructor result
`raceId`	Foreign key to `races.csv`
`constructorId`	Foreign key to `constructors.csv`
`points`	Points scored by the constructor in the race
`status`	Status of the constructor in the race

2.3.3 `constructor_standings.csv`

This file contains the constructor standings after each race.

Column	Description
`constructorStandingsId`	Unique identifier for each constructor standing
`raceId`	Foreign key to `races.csv`
`constructorId`	Foreign key to `constructors.csv`
`points`	Total points for the constructor after the race
`position`	Constructor's position in the standings
`positionText`	Text representation of the constructor's position
`wins`	Number of wins for the constructor in the season

2.3.4 `constructors.csv`

This file contains information about each constructor.

Column	Description
`constructorId`	Unique identifier for each constructor
`constructorRef`	A short reference name for the constructor
`name`	The full name of the constructor
`nationality`	The nationality of the constructor
`url`	URL for the constructor's Wikipedia page

2.3.5 `driver_standings.csv`

This file contains the driver standings after each race.

Column	Description
`driverStandingsId`	Unique identifier for each driver standing
`raceId`	Foreign key to `races.csv`
`driverId`	Foreign key to `drivers.csv`
`points`	Total points for the driver after the race
`position`	Driver's position in the standings
`positionText`	Text representation of the driver's position
`wins`	Number of wins for the driver in the season

DRAFT

2.3.6 `drivers.csv`

This file contains information about each driver.

Column	Description
`driverId`	Unique identifier for each driver
`driverRef`	A short reference name for the driver
`number`	The driver's car number
`code`	A three-letter code for the driver
`forename`	The driver's first name
`surname`	The driver's last name
`dob`	The driver's date of birth
`nationality`	The driver's nationality
`url`	URL for the driver's Wikipedia page

2.3.7 `races.csv`

This file contains information about each race.

Column	Description
`raceId`	Unique identifier for each race
`year`	The year of the race
`round`	The round number of the race in the season
`circuitId`	Foreign key to `circuits.csv`
`name`	The name of the Grand Prix
`date`	The date of the race
`time`	The time of the race
`url`	URL for the race's Wikipedia page
`fp1_date`	Date of Free Practice 1
`fp1_time`	Time of Free Practice 1
`fp2_date`	Date of Free Practice 2
`fp2_time`	Time of Free Practice 2
`fp3_date`	Date of Free Practice 3
`fp3_time`	Time of Free Practice 3
`quali_date`	Date of Qualifying
`quali_time`	Time of Qualifying
`sprint_date`	Date of Sprint Race
`sprint_time`	Time of Sprint Race

2.3.8 `results.csv`

This file contains the results for each driver in each race.

Column	Description
`resultId`	Unique identifier for each result
`raceId`	Foreign key to `races.csv`
`driverId`	Foreign key to `drivers.csv`
`constructorId`	Foreign key to `constructors.csv`
`number`	The driver's car number
`grid`	The driver's starting grid position

DRAFT

Column	Description
`position`	The driver's finishing position
`positionText`	Text representation of the driver's finishing position
`positionOrder`	The driver's finishing position in order
`points`	Points scored by the driver in the race
`laps`	Number of laps completed by the driver
`time`	The driver's total race time
`milliseconds`	The driver's total race time in milliseconds
`fastestLap`	The lap number of the driver's fastest lap
`rank`	The rank of the driver's fastest lap
`fastestLapTime`	The driver's fastest lap time
`fastestLapSpeed`	The speed of the driver's fastest lap
`statusId`	Foreign key to `status.csv`

2.3.9 `status.csv`

This file contains information about the finishing status of a driver in a race.

Column	Description
`statusId`	Unique identifier for each status
`status`	A description of the finishing status

2.4 Entity Relationship Diagram Model

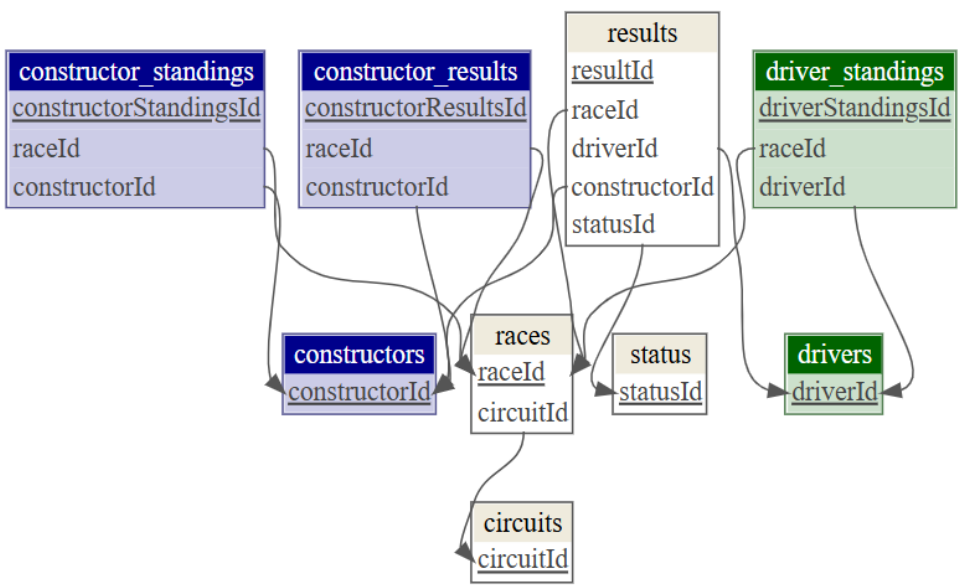

Figure 2: Entity Relationship Diagram Model

3. Data Transformation

This chapter is about the data transformation.

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9. References

[1] Hergé. 1956. “The Calculus Affair” https://www.tintin.com/en/albums/the-calculus-affair

[1] Jean-Luc. 2019. “Tintin – The Calculus Affair – The liberties of English translation” https://tintinomania.com/tintin-affaire-tournesol-anglais-brabham

[2] Kaggle “Formula 1 World Championship (1950 - 2024)” https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020

[3] Tracing Insights “Race Data” https://github.com/TracingInsights/RaceData

[4] Dataset copied to GitHub. Available at: https://github.com/jmr-lab/Formula-1 (Accessed: 2026).

[3] Rafael Irizarry. “Introduction to Data Science.” https://rafalab.dfci.harvard.edu/dsbook/

[8] Cover image from SVG SILH https://svgsilh.com/c3c3c3/image/309862.html

[9] Icons from Font Awesome https://fontawesome.com/

[10] The text of this report was reviewed by DuckAI, an AI language model that also assisted in reviewing and correcting some of my code : https://duck.ai/

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